Questions

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1. TCS NQT "Favorites" (Math, Number Theory & Logic)

These are extremely common in TCS NQT but rarer in standard coding interviews. You must know the logic to implement these from scratch (using modulo % and division / operators).

- **Number Properties (Implementation):**
  - Armstrong Number (Check if sum of cubes of digits = number)
  - Strong Number (Check if sum of factorials of digits = number)
  - Perfect Number (LeetCode 507)
  - Harshad Number (Number divisible by sum of its digits)
  - Abundant Number
  - Automorphic Number
  - Palindrome Number (LeetCode 9)
  - Reverse Integer (LeetCode 7)
- **Basic Math & Primes:**
  - GCD of Two Numbers (Greatest Common Divisor)
  - LCM of Two Numbers
  - Prime Numbers in a range / Count Primes (LeetCode 204)
  - Check if Prime Factors (LeetCode 2521 variant)
  - Factorial of a Number (Recursive & Iterative)
  - Fibonacci Number (LeetCode 509)
- **Conversions & Systems:**
  - Convert Binary to Decimal
  - Convert Decimal to Binary
  - Convert Octal to Decimal
  - Leap Year Check
- **Powers & Roots:**
  - Power of Two (LeetCode 231)
  - Power of Three (LeetCode 326) / Power of x, n (Medium)
  - Square Root of Integer (LeetCode 69)
  - Add Binary (LeetCode 67)
  - Count the Digits That Divide a Number
  - Minimum Cuts to Divide a Circle

2. Arrays (Logic & Pointers)

High frequency for both NQT and Digital roles.

- **Easy / Fundamentals:**
  - Two Sum (LeetCode 1)
  - Contains Duplicate (LeetCode 217)
  - Remove Duplicates from Sorted Array (LeetCode 26)

- Merge Sorted Array (LeetCode 88)
- Move Zeroes (LeetCode 283)
- Plus One (LeetCode 66)
- Best Time to Buy and Sell Stock (LeetCode 121)
- Majority Element (LeetCode 169)
- Single Number (LeetCode 136)
- Missing Number (LeetCode 268)
- Find the Middle Index in Array
- Pascal's Triangle (LeetCode 118)
- Running Sum of 1d Array (LeetCode 1480)
- Find All Numbers Disappeared in an Array (LeetCode 448)
- Summary Ranges (LeetCode 228)
- Intersection of Two Arrays (LeetCode 349) & Intersection II (LeetCode 350)

- **Medium / Optimization:**
  - Rotate Array (LeetCode 189)
  - Product of Array Except Self (LeetCode 238)
  - Maximum Subarray (Kadane's Algorithm) (LeetCode 53)
  - Maximum Ascending Subarray Sum
  - Sort Colors (0s, 1s, 2s) (LeetCode 75)
  - Find the Duplicate Number (LeetCode 287)
  - Container With Most Water (LeetCode 11)
  - 3Sum (LeetCode 15)
  - Max Consecutive Ones III
  - Largest Number
  - Count Array Pairs Divisible by K (Hard)
- **Hard / TCS Digital:**
  - Trapping Rain Water (LeetCode 42)
  - Median of Two Sorted Arrays (LeetCode 4)
  - Max Points on a Line

3. Strings (Parsing & Manipulation)

TCS loves string parsing problems where you cannot rely on built-in library functions.

- **Basic Manipulation:**
  - Reverse String (LeetCode 344)
  - First Unique Character in a String (LeetCode 387)
  - Valid Anagram (LeetCode 242)
  - Valid Palindrome (LeetCode 125)
  - Longest Common Prefix (LeetCode 14)
  - Length of Last Word (LeetCode 58)
  - Ransom Note (LeetCode 383)
  - Add Strings (LeetCode 415)
  - Multiply Strings (LeetCode 43)

    - Isomorphic Strings (LeetCode 205)
    - Word Pattern (LeetCode 290)
- **Advanced / Parsing:**
    - String to Integer (atoi) (LeetCode 8)
    - Implement strStr() (LeetCode 28)
    - Longest Substring Without Repeating Characters (LeetCode 3)
    - Group Anagrams (LeetCode 49)
    - Valid Parentheses (LeetCode 20)
    - Longest Palindromic Substring (LeetCode 5)
    - Count and Say (LeetCode 38)
    - Reverse Words in a String (LeetCode 151)
    - Reverse Words in a String III (LeetCode 557)
    - Roman to Integer (LeetCode 13)
    - Integer to Roman (LeetCode 12)
    - Simplify Path (Medium)
    - Palindrome Partitioning IV (Hard)

4. Sorting & Searching

Often asked to implement manual sorts (Bubble/Selection) rather than using .sort().

- **Searching:**
    - Binary Search (LeetCode 704)
    - First Bad Version (LeetCode 278)
    - Search Insert Position (LeetCode 35)
    - Search in Rotated Sorted Array (LeetCode 33)
    - Find First and Last Position of Element (LeetCode 34)
    - Single Element in a Sorted Array
    - Valid Perfect Square (LeetCode 367)
    - Arranging Coins (LeetCode 441)
- **Sorting & Intervals:**
    - **Manual Implementations:** Bubble Sort, Selection Sort, Insertion Sort.
    - Merge Intervals (LeetCode 56)
    - Kth Largest Element in an Array (LeetCode 215)
    - Excel Sheet Column Number (LeetCode 171)
    - Excel Sheet Column Title (LeetCode 168)

5. Linked Lists

Focus on pointer manipulation.

- **Standard Problems:**
    - Delete Node in a Linked List (LeetCode 237)
    - Reverse Linked List (LeetCode 206)
    - Merge Two Sorted Lists (LeetCode 21)
    - Linked List Cycle (LeetCode 141)
    - Middle of the Linked List (LeetCode 876)

- Palindrome Linked List (LeetCode 234)
- Remove Nth Node From End of List (LeetCode 19)
- Intersection of Two Linked Lists (LeetCode 160)
- Add Two Numbers (LeetCode 2)

6. Trees & Graphs

Basic Traversals and Properties.

- **Trees:**
  - Maximum Depth of Binary Tree (LeetCode 104)
  - Symmetric Tree (LeetCode 101)
  - Invert Binary Tree (LeetCode 226)
  - Same Tree (LeetCode 100)
  - Convert Sorted Array to Binary Search Tree (LeetCode 108)
  - Binary Tree Level Order Traversal (LeetCode 102)
  - Validate Binary Search Tree (LeetCode 98)
- **Graphs:**
  - Number of Islands (LeetCode 200)

7. Dynamic Programming (DP) & Others

For TCS Digital/Innovator roles.

- **Dynamic Programming:**
  - Climbing Stairs (LeetCode 70)
  - Min Cost Climbing Stairs (LeetCode 746)
  - House Robber (LeetCode 198)
  - Coin Change (LeetCode 322) & Coin Change II
  - Longest Increasing Subsequence (LeetCode 300)
  - Minimum Number of Refueling Stops (Hard)
- **Backtracking/Recursion:**
  - Permutations (LeetCode 46)
  - Subsets (LeetCode 78)

Patterns

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1. Two Pointers (Collision & Parallel)
2. Sliding Window
3. Fast & Slow Pointers (Tortoise & Hare)
4. Math & Number Theory (TCS Specials)
5. Hash Maps (Frequency & Lookup)
6. Binary Search
7. Stack (LIFO)
8. Dynamic Programming (DP)
9. BFS / DFS (Traversals)
10. Bit Manipulation (XOR)
11. Backtracking
12. Merge Intervals
13. Kadane's Algorithm (Greedy)
14. Prefix Sums (The Missing Pattern)
15. Heaps / QuickSelect (The Missing Pattern)

Patterns with Questions

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1. Two Pointers (Collision & Parallel)

Usage: Sorting, Pairs, Swapping, Reversing.

- Two Sum II (Input Array Is Sorted)
- Container With Most Water
- 3Sum
- Trapping Rain Water (Can also be DP/Stack)
- Remove Duplicates from Sorted Array
- Move Zeroes
- Reverse String
- Valid Palindrome
- Merge Sorted Array
- Intersection of Two Arrays / Intersection II
- Reverse Words in a String / Reverse Words III
- Sort Colors (Dutch National Flag)

2. Sliding Window

Usage: Subarrays, Substrings, Windows.

- Longest Substring Without Repeating Characters
- Max Consecutive Ones III
- Find All Anagrams in a String (implied by Anagrams)
- Minimum Size Subarray Sum (implied)

3. Fast & Slow Pointers (Tortoise & Hare)

Usage: Cycles, Middle of List, Duplicates.

- Linked List Cycle
- Middle of the Linked List
- Find the Duplicate Number
- Palindrome Linked List (Find middle, reverse second half)
- Happy Number (Cycle detection in digits)

4. Math & Number Theory (TCS Specials)

Usage: Digits, Primes, Divisors, Base Conversion.

- **Digit Logic:** Reverse Integer, Palindrome Number, Armstrong Number, Strong Number, Perfect Number, Harshad Number.
- **Primes/GCD:** Count Primes, GCD of Two Numbers, LCM, Prime Factors.
- **Conversions:** Binary to Decimal, Decimal to Binary, Octal to Decimal, Excel Sheet Column Number/Title.
- **Simple Math:** Plus One, Add Binary, Power of Two/Three, Factorial, Fibonacci Number (Iterative).
- **Geometry:** Minimum Cuts to Divide a Circle, Max Points on a Line.

5. Hash Maps (Frequency & Lookup)

Usage: Unsorted arrays, counts, duplicates.

- Two Sum (Unsorted)
- Contains Duplicate
- Majority Element
- Valid Anagram / Group Anagrams
- Ransom Note
- First Unique Character in a String
- Word Pattern / Isomorphic Strings
- Longest Common Prefix (Vertical scanning, but Hash Map can optimize).

6. Binary Search

Usage: O(log N) search in sorted data.

- Binary Search
- Search Insert Position
- First Bad Version
- Search in Rotated Sorted Array
- Find First and Last Position of Element
- Valid Perfect Square / Sqrt(x)
- Arranging Coins

7. Stack (LIFO)

Usage: Nested structures, Matching, Reverse Polish.

- Valid Parentheses
- Simplify Path
- Backspace String Compare (implied)

8. Dynamic Programming (DP)

Usage: Optimization, Ways to do X, Previous Results.

- Climbing Stairs / Min Cost Climbing Stairs
- House Robber
- Coin Change / Coin Change II
- Pascal's Triangle
- Longest Increasing Subsequence
- Maximum Ascending Subarray Sum
- Minimum Number of Refueling Stops (Hard DP)
- Count the Digits That Divide a Number
- Trapping Rain Water (DP approach)

9. BFS / DFS (Traversals)

Usage: Trees, Graphs, Islands.

- **Trees:** Maximum Depth of Binary Tree, Symmetric Tree, Invert Binary Tree, Same Tree, Validate BST, Convert Sorted Array to BST, Level Order Traversal.
- **Graphs:** Number of Islands.

10. Bit Manipulation (XOR)

Usage: Binary tricks.

- Single Number (XOR)
- Missing Number (XOR or Math)
- Number of 1 Bits (Hamming Weight) (Implied in conversion problems).

11. Backtracking

Usage: Generate all possibilities.

- Permutations
- Subsets
- Palindrome Partitioning IV

12. Merge Intervals

Usage: Overlapping time slots.

- Merge Intervals

13. Kadane's Algorithm (Greedy)

Usage: Max contiguous subarray.

- Maximum Subarray
- Best Time to Buy and Sell Stock (One-pass variation).

14. Prefix Sums (The Missing Pattern)

Usage: Range sums, Products without self.

- **Product of Array Except Self:** (Left product array + Right product array).
- **Running Sum of 1d Array.**
- **Find Pivot Index / Middle Index.**

15. Heaps / QuickSelect (The Missing Pattern)

Usage: "Top K", "Kth Largest/Smallest".

- **Kth Largest Element in an Array:** Use a Min-Heap (size k) or QuickSelect.
- **Merge K Sorted Lists:** (Not in your list, but commonly paired here).